

Use the information below to answer questions 1 and 2:

A wire of length 12 cm is moved at right angles across a magnetic field at a constant velocity of 2 metres per second. The flux density in the magnetic field is 20 mWb/m².

1. Find the e.m.f. generated in millivolts.

A 6.7
B 5.9
C 4.8
D 3.2

solution

$$L = 12 \times 10^{-2} \text{ m}$$

$$v = 2 \text{ m/s}$$

$$B = 20 \text{ mWb/m}^2$$

$$E = BLv$$

$$= 12 \times 10^{-2} \times 2 \times 20$$

$$= 4.8 \text{ mV}$$

2. What force in newtons will be required to move this conductor if the e.m.f. causes a current of 10 A to flow in an external circuit?

A 0.053
B 0.045
C 0.037
D 0.024

solution

$$B = 20 \times 10^{-3} \text{ Wb/m}^2$$

$$I = 10 \text{ A}$$

$$L = 12 \times 10^{-2} \text{ m}$$

$$\theta = 90$$

$$F = BIL \sin \theta$$

$$= 20 \times 10^{-3} \times 10 \times 12 \times 10^{-2}$$

$$= 0.024 \text{ N}$$

Use the information below to answer questions 3 and 4:

A 600-V dc motor has an armature circuit resistance of 0.14 ohms. At no load, when running at 800 rev/min, the armature current is 12 A. The full-load armature current is 225 A.

3. Find the full-load speed in rev/min when the field pole flux is constant.

A 840
B 800
C 760
D 720

solution

$$V = 600$$

$$R_a = 0.14 \Omega$$

$$N_1 = 800 \text{ rev/min}$$

$$I_a = 12 \text{ A, No load}$$

$$\phi = \text{constant}$$

$$E_1 = V - I_a R_a$$

$$= 600 - (12 \times 0.14)$$

$$= 598.32 \text{ V}$$

On full load,

$$I_a = 225 \text{ A}$$

$$E_2 = V - I_a R_a$$

$$= 600 - (225 \times 0.14)$$

$$= 568.5 \text{ V}$$

but $E = k\omega\phi$, $\phi = \text{constant}$

$$\Rightarrow E = k\omega$$

$$\frac{E_1}{N_1} = \frac{E_2}{N_2}$$

$$\Rightarrow N_2 = \frac{568.5 \times 800}{598.32}$$

$$= 760 \text{ rev/min}$$

4. Find the full-load speed in rev/min when the field pole flux is reduced to 80% of its original value.

A 1200
B 1150
C 1050
D 950

$\phi_2 = 0.8\phi_1$

$$E = k\omega\phi$$

$$E_1 = E_2$$

$$N_1\phi_1 = N_2\phi_2$$

$$N_2 = \frac{E_2 N_1 \phi_1}{E_1 \phi_2} = \frac{568.5 \times 800 \times \phi_1}{598.32 \times 0.8\phi_1}$$

$$= 950 \text{ rev/min}$$

Use the information below to answer questions 5 and 6:

A 125-V dc shunt motor has an armature circuit resistance of 0.2 ohms and a field resistance of 45 ohms. The motor draws a line current of 50 A.

5. Find the back e.m.f. of the motor in volts

A 118
B 116
C 114
D 112

Handwritten solution for question 5:

$$\begin{aligned} \text{C-)} V &= 125V ; R_f = 45\Omega \\ R_a &= 0.2\Omega ; I_L = 50A \\ I_f &= \frac{125}{45} = 2.77A \\ I_a &= I_L - I_f ; E = V - I_a R_a \\ &= 50 - 2.77 \\ &= 47.23A \\ E &= 125 - (47.23 \times 0.2) \\ &= 115.5, \approx 116V \end{aligned}$$

6. Find the mechanical power developed by the motor in kilowatts.

A 7.84
B 6.56
C 5.46
D 4.35

Handwritten solution for question 6:

$$\begin{aligned} M_p &= E I_a \\ &= 115.5 \times 47.2 \\ &= 5.46 \text{ kW} \end{aligned}$$

Use the information below to answer questions 7 and 8:

A series motor has combined series field and armature resistance of 0.85 ohms, runs at 1,000 rev/min, and draws 20 A from a 250-V source

7. Find the developed torque in N.m.

A 44.50
B 42.24
C 40.80
D 38.42

Handwritten solution for question 7:

$$\begin{aligned} \text{Solution} \\ R_T &= 0.85\Omega ; I_f = 20A \\ N &= 1000 \text{ rev/min} ; V = 250V \\ P &= I_a E - \text{losses} \\ \text{but } E &= V - I_a R_T \\ &= 250 - (20 \times 0.85) \\ &= 233V \\ E I_a &= \frac{2\pi N T}{60} \\ \Rightarrow T &= \frac{60 E I_a}{2\pi N} = \frac{60 \times 233 \times 20}{2\pi \times 1000} \\ &= 44.50 \text{ N.m} \end{aligned}$$

8. Find the speed in rev/min if a 3.75-ohm resistor is connected in series with the motor and it draws 20 A from a 250-V source.

A 710
B 678
C 540
D 403

Handwritten solution for question 8:

$$\begin{aligned} \text{Solution} \\ R_{T2} &= 0.85 + 3.75 \\ &= 4.6 \\ E_2 &= V - I_a R_{T2} \\ &= 250 - (20 \times 4.6) \\ &= 158, \text{ since } V, I_a \text{ is constant, } \phi = k \\ \Rightarrow N_2 &= \frac{E_2 N_1}{E_1} = \frac{158 \times 1000}{233} \\ &= 678.11, \approx 678 \text{ rev/min} \end{aligned}$$

Use the information below to answer questions 9 to 12

An eight-pole armature is wound with 480 conductors. The magnetic flux and the speed are such that the average e.m.f. generated in each conductor is 2.2 V, and each conductor is capable of carrying a full-load current of 100 A.

9. Find the terminal voltage on no load in volts if the armature is lap-connected.

- A 162
- B 153
- C 144
- D 132

$N_p = \frac{N}{2}$ pole = 8 ; $N_s = \text{conductors}$ per pole
 $N_s \text{ conductors} = 480$
 $E_c = 2.2 \text{ V}$
 $I_f = 100 \text{ A}$
 in lap, $a = P$
 $\therefore E = \frac{2 N P \phi}{60 a} = \frac{N_s E_c}{a}$
 $N_s = \frac{480}{8} = 60$
 $\therefore E = 60 \times 2.2 = 132 \text{ V}$

10. Find the output current on full load in amperes if the armature is lap-connected.

- A 800
- B 750
- C 700
- D 650

Output current = $I_f \times N_p$
 $= 100 \times 8$
 $= 800 \text{ A}$

11. Find the terminal voltage on no load in volts if the armature is wave-connected.

- A 556
- B 544
- C 528
- D 506

In wave, $a = 2$, number of parallel paths
 $\Rightarrow N_s = \frac{480}{2} = 240$
 $\therefore E = E_c N_s$
 $= 240 \times 2.2$
 $= 528 \text{ V}$

12. Find the output current on full load in amperes if the armature is wave-connected.

- A 200
- B 180
- C 160
- D 140

Output current = $I_f \times a$
 $= 100 \times 2$
 $= 200 \text{ A}$